DDG REGISTRATION DESK.

OrbitBot Galaxy launch · Kraftverket, Oslo · 3 Dec 2026 · ~150 guests, one every ~18 s at peak

Stack · Steps · Components

THE STACK.

FRONTEND
Next.js 16 + React 19
App Router, TypeScript, Tailwind v4 with the DDG UI kit

BACKEND
Next.js API routes
Server-only service layer, PIN sessions in signed cookies

DATABASE
Supabase Postgres
EU (Ireland), Row-Level Security on, no public access

HOSTING
Vercel
Live URL, env secrets, deploy from GitHub

LIBRARIES
Excel · phone · QR
exceljs, libphonenumber-js, qrcode, qr-scanner

QUALITY
Vitest + DB check
Service and import tests, 10 parallel check-ins → 1 admitted

HOW IT FITS TOGETHER.

DEVICES (BROWSER)

Desk phones
Search, scan QR, check in

Manager laptop
Arrivals, review, report

Guest phone · Venue screen
Ticket QR · Galaxy wall

Never talks to the database directly

NEXT.JS ON VERCEL (SERVER)

Pages
/desk · /manager/* · /t/[token] · /wall/[key]

API routes
/api/auth · /api/desk · /api/manager · /api/wall

Gatekeeper
requireRole("desk" | "manager"), PIN rate limit, audit every action

Services
checkin · import · duplicates · walkins · alerts · report · wall

Store interface
One contract for all data access: Supabase in production, in-memory store in tests

SUPABASE POSTGRES (EU)

Secret key, server only
RLS on every table, zero public policies

Unique index
One active check-in per person. The database blocks double admission.

Original rows kept
Raw + cleaned values side by side, nothing auto-merged

EVENT DAY, 5 STEPS.

1. **Upload Excel** — Keep every row, clean key fields
2. **Review duplicates** — Manager confirms or rejects
3. **Guest arrives** — Scan QR or search name
4. **Check in** — Once per person, live counter
5. **Report** — .xlsx for the client

HOW IT WAS BUILT.

1. **Grill** — Q&A on scope, privacy, data
2. **Spec** — SPEC.md, Must / Next / Later
3. **Tickets** — 13 vertical slices on GitHub
4. **Contracts** — Types, schema, store, check-in
5. **Agent team** — Parallel slices, own files
6. **Verify** — Tests, DB race, phone, deploy

COMPONENTS.

Four areas, one app. Every area has its own access level.

What the app contains

Desk

`/desk · desk PIN`

- Big search (ignores accents and case)
- QR scanner button
- Guest card + 72px CHECK IN
- Welcome message + Undo
- VIP / wheelchair banners
- Call manager button
- Walk-in form
- Live counter

Manager

`/manager · manager PIN`

- Live Arrivals + alerts
- Guests: raw vs cleaned
- Walk-in approvals
- Duplicate review
- Tickets: QR + CSV links
- Excel import
- Client report (.xlsx)
- Do-not-admit override

Guest ticket

`/t/[token] · public link`

- Personal QR code
- Random 128-bit token
- No personal data in the QR
- Can be reissued

Galaxy wall

`/wall/[key] · secret link`

- Big-screen animation
- Arriving companies
- Live count
- No guest names

DATABASE TABLES.

imports
Each Excel upload

registrations
Raw row + cleaned fields

people
One real person

duplicate_suggestions
Strong / medium / weak

checkins
Max 1 active per person

walkins
Wait for approval

alerts
VIP, access, help calls

audit_log
Every action

login_attempts
PIN rate limit

Many registrations can point to one person. Check-in is per person, so a confirmed duplicate is admitted once.

CODE BUILDING BLOCKS.

`lib/domain`	Shared types (the contract)
`lib/import`	Parse, clean, flag, suggest duplicates
`lib/service`	Check-in, walk-ins, alerts, report, wall
`lib/server/store`	Supabase + in-memory store
`lib/server/session`	PIN roles, signed cookie
`components/ui`	DDG design kit
`components/desk · qr`	Desk screen, scanner, QR
`db/`	Schema, migrate, reset, race test

BUILT-IN GUARANTEES.

1. **Never admit twice**
   Enforced by the database, even with two desks at once
2. **Desk sees the minimum**
   No phone, email, raw rows or manager notes
3. **Nothing is lost**
   Original rows kept, merges only by a manager
4. **Accurate client numbers**
   Duplicates once, staff apart, exceptions listed
5. **Private by default**
   EU data, no guest file in git, full audit trail

Next: QR in confirmation email · self check-in kiosk · badge printing · safe re-import